

Name: Abdullah Ayub

Roll NO: BSDSF24M029

Ex: Find Volume of Region D enclosed by surfaces:

$$z = x^2 + 3y^2 \text{ and } z = 8 - x^2 - y^2$$

$$x^2 + 3y^2 = 8 - x^2 - y^2$$

$$2x^2 + 4y^2 - 8 = 0 \quad \text{---(i)}$$

$$4y^2 = 8 - 2x^2$$

$$y = \pm \sqrt{\frac{8 - 2x^2}{4}} = \pm \sqrt{\frac{4 - x^2}{2}}$$

Put $y=0$ in (i) for limit

$$2x^2 = 8$$

$$x^2 = 4$$

$$x = \pm 2$$

$$V = \iiint_D dz dy dx = \int_{-2}^{+2} \int_{-\sqrt{\frac{4-x^2}{2}}}^{+\sqrt{\frac{4-x^2}{2}}} \int_{x^2+3y^2}^{8-x^2-y^2} dz dy dx$$
$$= \int_{-2}^{+2} \int_{-\sqrt{\frac{4-x^2}{2}}}^{+\sqrt{\frac{4-x^2}{2}}} z \Big|_{x^2+3y^2}^{8-x^2-y^2} dy dx$$

$$= \int_{-2}^{+2} \int_{-\sqrt{\frac{4-x^2}{2}}}^{+\sqrt{\frac{4-x^2}{2}}} (8 - x^2 - y^2 - x^2 - 3y^2) dy dx = \int_{-2}^{+2} \int_{-\sqrt{\frac{4-x^2}{2}}}^{+\sqrt{\frac{4-x^2}{2}}} (8 - 2x^2 - 4y^2) dy dx$$

$$= \int_{-2}^{+2} \left[8y - 2x^2y - \frac{4y^3}{3} \right]_{-\sqrt{\frac{4-x^2}{2}}}^{+\sqrt{\frac{4-x^2}{2}}} dx = \int_{-2}^{+2} \left((8 - 2x^2)y - \frac{4}{3}y^3 \right) \Big|_{-\sqrt{\frac{4-x^2}{2}}}^{+\sqrt{\frac{4-x^2}{2}}} dx$$

$$= \int_{-2}^{+2} \left((8 - 2x^2) \sqrt{\frac{4-x^2}{2}} + (8 - 2x^2) \sqrt{\frac{4-x^2}{2}} + \left(\frac{-4}{3} \left(\frac{4-x^2}{2} \right)^{3/2} - \frac{4}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right) \right) dx$$

$$= \int_{-2}^{+2} \left(2(8 - 2x^2) \sqrt{\frac{4-x^2}{2}} - \frac{8}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right) dx$$

$$= \int_{-2}^{+2} \left(4(4-x^2) \sqrt{\frac{4-x^2}{2}} - \frac{8}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right) dx$$

$$\sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2}$$

$$4^{3/2} = (2^2)^{3/2} = 2^3 = 8$$

$$\frac{24-8}{3}$$

$$= \int_{-2}^{+2} \left(\frac{4 \times 2 \times (4-x^2)}{2} \times \sqrt{\frac{4-x^2}{2}} - \frac{8}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right) dx$$

$$= \int_{-2}^{+2} \left(8 \cdot \left(\frac{4-x^2}{2} \right)^{3/2} - \frac{8}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right) dx$$

$$= \frac{16}{3} \int_{-2}^{+2} \frac{(4-x^2)^{3/2}}{\sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2}} dx$$

$$= \frac{\sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2} \times 4}{3 \sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2}} \int_{-2}^{+2} (4-x^2)^{3/2} dx$$

$$= \frac{4\sqrt{2}}{3} \int_{-2}^{+2} (4-x^2)^{3/2} dx \quad \text{--- (a)}$$

$$\begin{cases} \text{Put } x = 2\sin\theta & x \rightarrow -2, \theta \rightarrow -\pi/2 \\ \frac{dx}{d\theta} = 2\cos\theta & x \rightarrow 2, \theta \rightarrow \pi/2 \end{cases}$$

$$= \int_{-\pi/2}^{\pi/2} (4-4\sin^2\theta)^{3/2} 2\cos\theta d\theta$$

$$= 2 \cdot 4 \int_{-\pi/2}^{\pi/2} (1-\sin^2\theta)^{3/2} \cos\theta d\theta$$

$$= 16 \int_{-\pi/2}^{\pi/2} (\cos^2\theta)^{3/2} \cos\theta d\theta = 16 \int_{-\pi/2}^{\pi/2} \cos^3\theta \cdot \cos\theta d\theta$$

$$= 16 \int_{-\pi/2}^{\pi/2} (\cos^2\theta)^2 d\theta = 16 \int_{-\pi/2}^{\pi/2} \left(\frac{1+\cos 2\theta}{2} \right)^2 d\theta$$

$$= \frac{16}{4} \int_{-\pi/2}^{\pi/2} (1+2\cos 2\theta+\cos^2 2\theta) d\theta = 4 \int_{-\pi/2}^{\pi/2} \left(1+2\cos 2\theta + \frac{1+\cos 4\theta}{2} \right) d\theta$$

$$= 2 \int_{-\pi/2}^{\pi/2} (2+4\cos 2\theta+1+\cos 4\theta) d\theta$$

$+\pi/2$

$$= 2 \int_{-\pi/2}^{\pi/2} (3 + 4\cos 2\theta + \cos 4\theta) d\theta$$

$$= 2 \left[3\theta + \frac{4\sin 2\theta}{2} + \frac{\sin 4\theta}{4} \right]_{-\pi/2}^{\pi/2}$$

$$= 2 \left[3\left(\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right) + 2(\sin \pi + \sin \pi) + \frac{1}{4}(\sin 2\pi + \sin 2\pi) \right]$$

$$= 2 \cdot 3 \cdot \pi + 0 + 0$$

$$= 6\pi \quad \text{--- (ii)}$$

Put (ii) in (i)

$$= \frac{4\sqrt{2} \cdot 6\pi}{2}$$

$$= \boxed{8\pi\sqrt{2}}$$